

Data Science

Data Science: - Study of data to extract meaningful insights for business.

* It can range from a simple array of a few numeric observations to a complex matrix of millions of observations with thousands of variables.

→ Extracting Meaningful Patterns

* One of the key aspects of data science is the process of generalization of patterns from a dataset.

→ Building Representative Models

* A model is the representation of a relationship between variables in a dataset.

* Modeling is a process in which a representative abstraction is built from the observed dataset.

→ Data Science is a combination of statistics, machine learning & computing.

Associated Fields with D.S

1) Descriptive statistics: - computing mean, standard deviation, correlation & other descriptive statistics, quantify the aggregate structure of a dataset.

Real world applications

- Business Analytics.
- Healthcare Informatics.
- Financial Modeling.
- Social Network Analysis.

Applications

- Image Recognition.
- Speech Recognition.
- Fraud & Risk Detection.
- Delivery Logistics.
- Gaming.
- Price Comparison Websites.
- Recommender systems.
- Digital Advertisements.
- Internet Search.

Tools & Technologies

- * Programming Languages ⇒ Python, R
- * Data Visualization Tools ⇒ Tableau, Power
- * M.L. Libraries ⇒ Scikit - learn, TensorFlow
- * Big Data Frameworks ⇒ Apache Hadoop & Spark.

2) Exploratory Visualization:- The process of expressing data in visual coordinates enables users to find patterns & relationships.

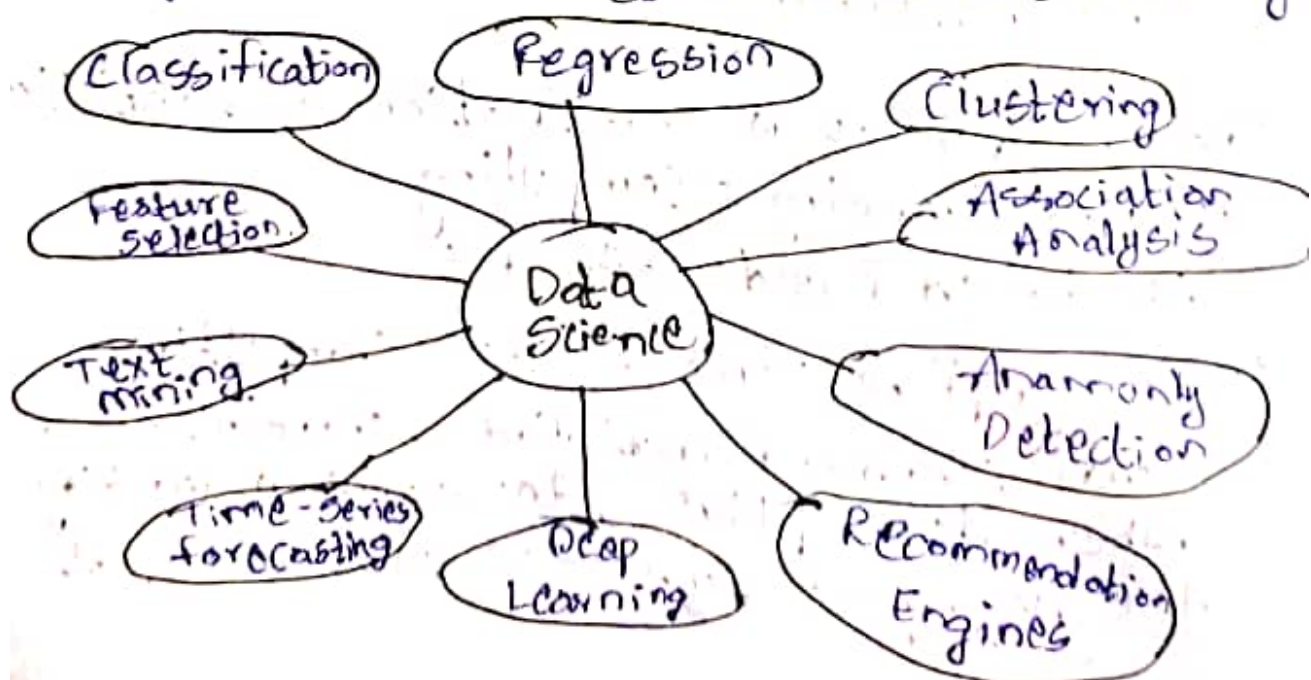
3) Dimensional Slicing:- Provide information on the data through dimensional slicing filtering & pivoting.

Ex:- OLAP.

4) Hypothesis testing:- Experimental data are collected to evaluate whether a hypothesis has enough evidence to be supported or not.

5) Data Engineering.

6) Business Intelligence:- This helps the organizations consume data effectively.



Data collection

- * Process of gathering & acquiring data from various sources.

Data cleaning (cleansing)

- * Process of detecting & correcting errors, inconsistencies & inaccuracies in datasets.

Exploratory Data Analysis

- * Process of analyzing data sets to summarize their main characteristics, with visual methods.

Predictive modeling

- * Process of using historical data to make predictions about future outcomes.

Data Visualization

- * Graphical representation of information & data.

Statistical Analysis

- * Process of collecting, exploring, summarizing & interpreting data to uncover patterns, trends & relationships & insights.

Decision making

- * Selecting the best course of action from available alternatives.

Advantages of CRISP-DM

- Industry - standard
- Flexible Framework
- Iterative process
- Easy Project Management
- Business - oriented Approach
- * Used in Industries like,
 - Healthcare, Finance, Retail, Manufacturing.

Limitations

- * Requires experienced professionals.
- * Time consuming for large projects
- * Limited Automation.

SEMMA

* Sample, Explore, Modify, Model, Assess

* To simplify data mining process.

Workflow

→ Select representative sample

↓
Explore relationships

↓
Modify variables

↓
Build models

↓
Assess performance.

Challenges for DS

- Data Quality.
- Data Privacy & Security.
- Scalability & Performance.
- Interpretability & Explainability.

DS Roles

- Data Scientist
- Data Engineer
- Business Analyst
- Data Architect
- BI Analyst
- AI Engineer
- NLP Engineer
- Statistician.
- M.L Engineer

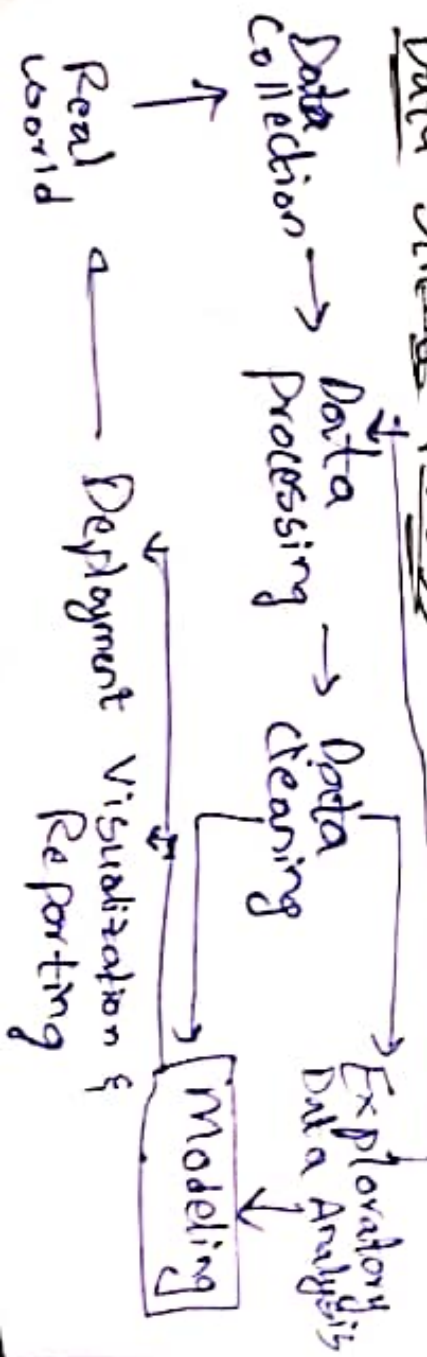
CRISP-DM

* Cross Industry Standard Process for Data Mining.

- Provides structured & systematic approach for solving business problem using data.
- Helps to improve project planning, data analysis, model deployment, & deployment.

	Data Science	B.I.
Concept	used to discover the hidden patterns in the data.	Used by the enterprise for business data analysis.
Focus	Focuses on the future	*Focuses on the past & present.
Data	Deals with both structured & un-structured data	Only structured.
Flexibility	More Flexible	Less Flexible.
Complexity	More Complex	Simpler
	Deals with what will happen & what if	Deals with what happened.
Storage	Real-time clusters	Data warehouse is utilized to hold data.
	ELT	ETL
Tools	SAS, BigML, MATLAB, Excel	Insight'squared, sales Analytics, Klipfolio, Thoughtspot, Cyfe, TIBCO, Spofire.

Data Science Process



Data types

1) Numeric Data types

* Integer

* Float

* Complex $\Rightarrow$ used in advanced mathematics

2) String

3) List $\Rightarrow$ Mutable

4) Tuple $\Rightarrow$ Immutable.

5) Set

6) Dictionary

7) Boolean

Pandas

* Python library used for data analysis cleaning & data manipulation.

Pandas Core Data Structures

1) Series (1D)

2) Dataframe (2D)

3) Index (Theoretical)

4) Series

* 1-Dimensional labeled array capable of holding,

- $\rightarrow$ Int
- $\rightarrow$ Float
- $\rightarrow$ Strings
- $\rightarrow$ Objects.

* Like a single column in Excel.

Phases of CRISP-DM

- 1) Business Understanding
- 2) Data Understanding
- 3) Data Preparation
- 4) Modeling
- 5) Evaluation
- 6) Deployment

1) Business Understanding

- * To understand organizational goals.

Ex: Predict customer churn in banking.

2) Data Understanding

- * Collect & understand data relevant to the problem.

Activities

→ Data Collection ⇒ Through surveys, sensors & IoT devices, web scraping

→ Exploratory Data Analysis To understand patterns & relationships, detect anomalies

→ Data Quality Verification

- * Missing values
- * Duplicate records
- * Noise
- * Inconsistent formats
- * Outliers

3) Data Preparation

- * Transform raw data into usable format.
- Data cleaning
- Data integration
- Feature Engineering.
- Data transformation:

4) Modeling

- * Build machine learning models for prediction or analysis.

→ Common Algorithms

- * Classification
 - Spam Detection
 - Disease Prediction
- * Regression
 - Price Prediction
 - Sales Forecasting.

* Clustering → Customer segmentation

* Recommendation Systems → Netflix

Recommendation

5) Evaluation

- * Ensures the model has satisfied business objectives or not.

6) Deployment

- * Deploy the model into real-world systems.

2) DataFrame

* A dataframe is a 2D labeled data structure.

→ Rows (index)

→ Columns (labels)

* A dataframe consists of multiple series.

Ex:- import pandas as pd

data = {

"Name": ["Ran", "Sai", "Anu"]

"Marks": [85, 90, 88]

}

df = pd.DataFrame(data)

print(df)

O/P:-

	Name	Marks
0	Ran	85
1	Sai	90
2	Anu	88

Characteristics

2) Heterogeneous Data ⇒ Each column have different types.

data = {

"Name": ["A", "B"],

"Age": [90, 25]}

Ex:- import pandas as pd

S = pd.Series([10, 20, 30], index = ['A',

print(S).

o/p:- A 10

B 20

C 30

Characteristics

1) Indexed Data $\Rightarrow$ print (S["A"])

o/p:- 10

2) Supports Multiple Data Types

Ex:- S = pd.Series([10, 30.5, "Hello"])

3) Automatic Indexing

Ex:- If indexes are not given, Pandas assigns $\Rightarrow 0, 1, 2, \dots$ automatically.

4) Vectorized Operations

S = pd.Series([1, 2, 3])

print (S*2)

o/p:- [2, 4, 6]

Used in

$\rightarrow$ Time-Series data

$\rightarrow$ Sensor readings

$\rightarrow$ Single feature Analysis.

$\rightarrow$ ~~Real-time~~ Real-time Sensor

→ Business-focused → Includes deployment → Technology independent → End-to-end lifecycle → Flexible	→ Data-focused → No deployment phase → SAS-oriented → Mainly data mining → Tool-specific
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Python

Features

- Easy to learn.
- Interpreted
- Dynamically typed (No need to declare variable types).
- Cross-platform (Runs on Windows, macOS & Linux)
- Large ecosystem.

Applications

- Web development (Django, Flask).
- Data science & analysis (Pandas & NumPy)
- ML & AI (TensorFlow, PyTorch)
- Automation & Scripting.
- Game Development
- Desktop applications.

4) IoT is sensor data $\Rightarrow$ Data collected from physical devices connected to the internet.

5) Databases $\Rightarrow$ a) Relational ? Database
b) NoSQL ? Database

c) Web data $\Rightarrow$ Info collected from websites, web applications & online platforms.

Data Pre-processing

* It is the process of transforming raw data into an understandable format.

Quality of the data can be checked with,

- $\rightarrow$ Accuracy
- $\rightarrow$ Completeness
- $\rightarrow$ Consistency
- $\rightarrow$ Timeliness
- $\rightarrow$ Believability
- $\rightarrow$ Interpretability.

* 4 major tasks in data pre-processing,

- 1) Data cleaning.
- 2) Data integration.

2) Column-based Operations $\Rightarrow$ `df["Marks"]`

3) Row-based operations $\Rightarrow$ `df.loc[]`
`df.iloc[]`

4) Size & Shape $\Rightarrow$ `df.shape`

5) Statistical Functions $\Rightarrow$ `df.describe()`,
`df.mean()`

* Dataframe allows,

$\rightarrow$ Data cleaning & analysis, filtering,
Aggregation.

3) Index

* Label system that identifies rows & columns.

* Supports alignment of data!

Note: - `df.head()` $\Rightarrow$ Displays first rows

`df.tail()` $\Rightarrow$ Displays last rows.

`df[df["Marks"] > 75]` $\Rightarrow$ Students scoring above 75.

`df["Marks"].mean()` $\Rightarrow$ Average Marks.

`df["Marks"].max()` $\Rightarrow$ Max marks.

`df["Marks"] = df["Marks"].fillna(df["Marks"].max)`

^{it} Handles missing values.

`df = df.sort_values("Marks")`

`df = pd.read_csv("Sales-data.csv")`

Data Collection

* Systematic: process of gathering, measuring, & analyzing information to extract insights & make informed decisions.

* Poor quality data leads to biased models & incorrect conclusions, so data collection is critical.

Types of Data Sources

- 1) Primary
- 2) Secondary
- 3) Real-time $\Rightarrow$ Continuously generated & collected from sources like IoT devices.

Methods of Data Collection

- 1) Surveys & Questionnaires
- 2) Web Scraping $\Rightarrow$ Process of automatically extracting data from websites.
- 3) APIs (Application Programming Interfaces)

* These provide a structured way to access data from various platforms.

Ex: Weather API, Google Maps API, Twitter API.

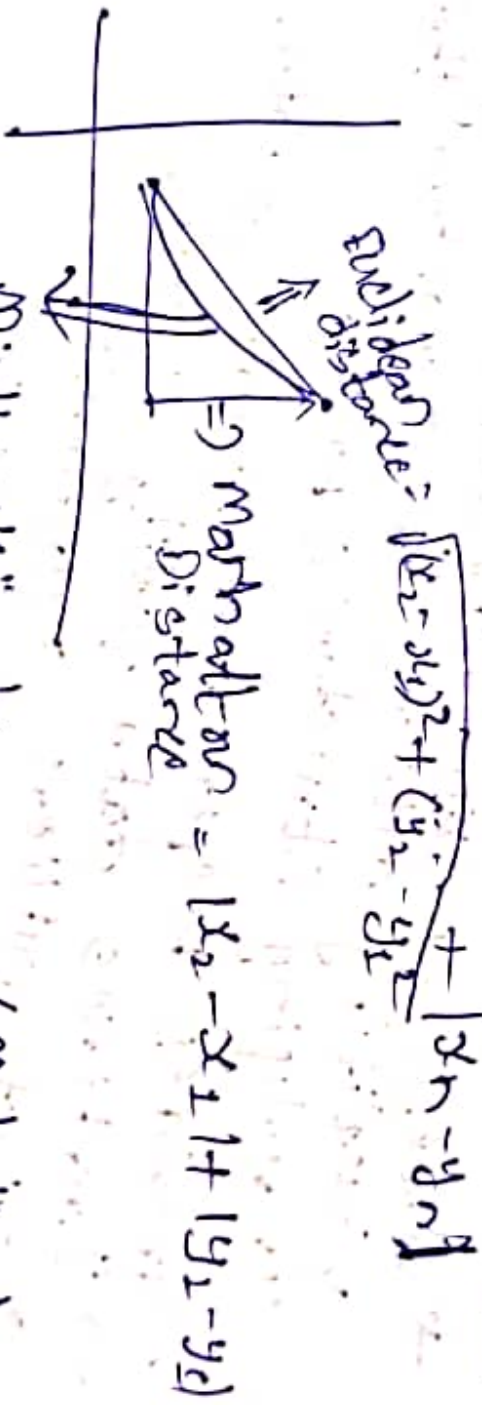
Common Technologies

→ Orchestration: Apache Airflow, Dagster, Prefect.

→ Processing: Apache Spark, Databricks
→ Storage: Google BigQuery, Snowflake, S3.

Distance Measures for Numerical Data

→ Euclidean Distance $\Rightarrow [(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2]$
→ Manhattan Distance $\Rightarrow |x_1 - y_1| + |x_2 - y_2| + \dots + |x_n - y_n|$



Minkowski when $p=1$ (Manhattan)
Minkowski when $p=2$ (Euclidean)

Similarity Measures

1) Cosine Similarity:

$$\cos \theta = \frac{A \cdot B}{\|A\| \cdot \|B\|}$$

2) Jaccard Similarity
 $= \frac{A \cap B}{A \cup B}$

- 3) Data Reduction
- iv) Data Transformation.

Data Issues

- 1) Missing Data
- 2) Duplicate Data
- 3) Inconsistent Data
- 4) Inaccurate Data
- 5) Outdated Data
- 6) Invalid Data
- 7) Redundant Data (Repetition of Data).

Handling Missing Data

- Binning Method
- Clustering
- Regression

Data Pipeline

* A data pipeline is a set of tools & processes for collecting, processing, & delivering data from one or more sources to a destination where it can be analyzed & used.

Overview

Data Extraction



Staging Area

Transformation Load



Transformation

Components

- Source
- Destination
- Data Flow
- Processing
- Workflow
- Monitoring

Pipeline Stages

Data Collection



Data Ingestion



Processing Transformation



Data Storage



Consumption

Pipeline Types

- ETL
- ELT